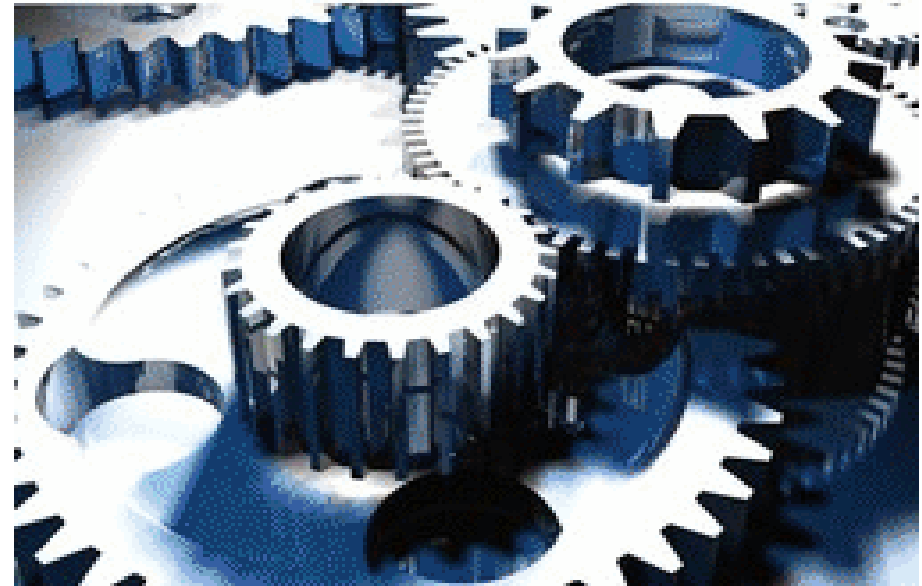


INTRODUCTION TO THE MODULE

ME1031 – Engineering Skill Development

Department of Mechanical Engineering, Sri Lanka
Institute of Information Technology



GIF courtesy: <https://gifer.com/en/83WL>

Presented by Dr. L U Subasinghe | PhD (SG), BSc Eng Hons (SL), CEng.MIE(SL), AMIMechE
Senior Lecturer

Department of Mechanical Engineering, University of Moratuwa
lihils@uom.lk

LEARNING OUTCOMES

- After completing this module, the students should be able to,
 - LO-1: Explain the importance of engineering drawings as a medium of communication to convey ideas in engineering fields.
 - LO-2: Draw orthographic projections and sectional views with dimensioning
 - LO-3: Draw Isometric views and surface development of solids
 - LO-4: Apply principles in manual drawings to create objects and their views using CAD software.
 - **LO-5: Practice safety guidelines and wearing of personal protective equipment in the engineering workshop**
 - **LO-6: Use of basic hand tools and machine tools in engineering workshop for machining processes and joining metals**

OUTLINE

Workshop Practice

- Introduction to basic manufacturing processes
- Introduction to basic machining process: Use of Hand tools, Power operated hand Tools, Machine Tools
- Measurement and basic measuring tools
- Introduction to basic joining processes: mechanical, metallurgical, bonding
- Workshop safety: Safety guidelines, Personal safety equipment, Fire safety
- Introduction to work materials and tool materials

MODULE WEIGHT

- GPA 3.0 credits
- Lectures: 28 hours
- Assessment Criteria
 - Written Exam: 40%
 - Continuous Assessments: 60%

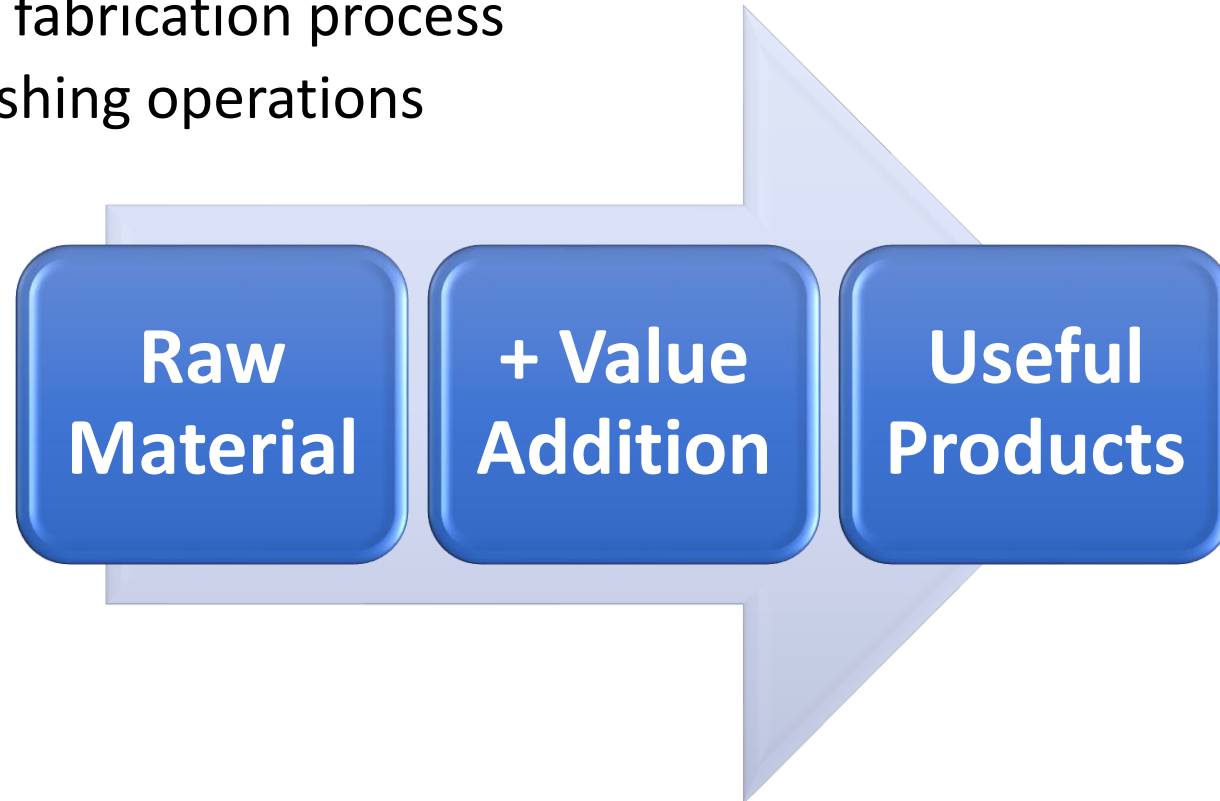
OUTLINE OF TODAY'S LECTURE

- Introduction to manufacturing
- Different types of manufacturing processes
 - Casting
 - Forming
 - Machining
 - Joining

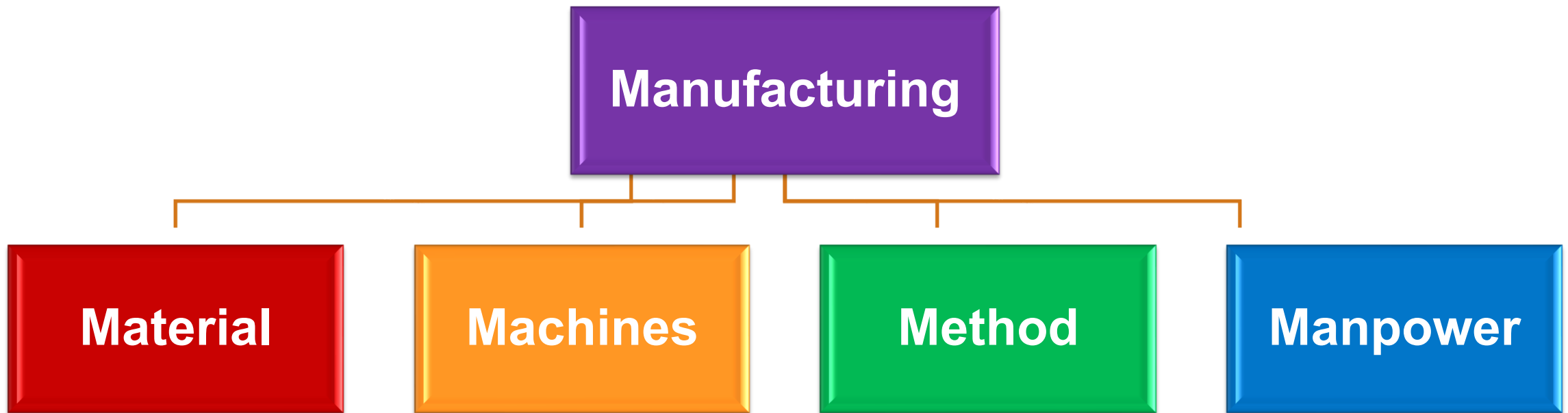
INTRODUCTION TO MANUFACTURING

WHAT IS MANUFACTURING

- It is a process of converting raw material into useful products
 - Designing the product
 - Selecting materials
 - Selecting tools
 - Performing the fabrication process
 - Performing finishing operations



MAJOR REQUIREMENTS OF MANUFACTURING



MANUFACTURED GOODS

- Two types of manufactured goods

Producer
goods



Consumer
goods



DIFFERENT TYPES OF MANUFACTURING PROCESSES

MANUFACTURING PROCESSES

Casting

Forming

Machining

Joining

CASTING

CASTING

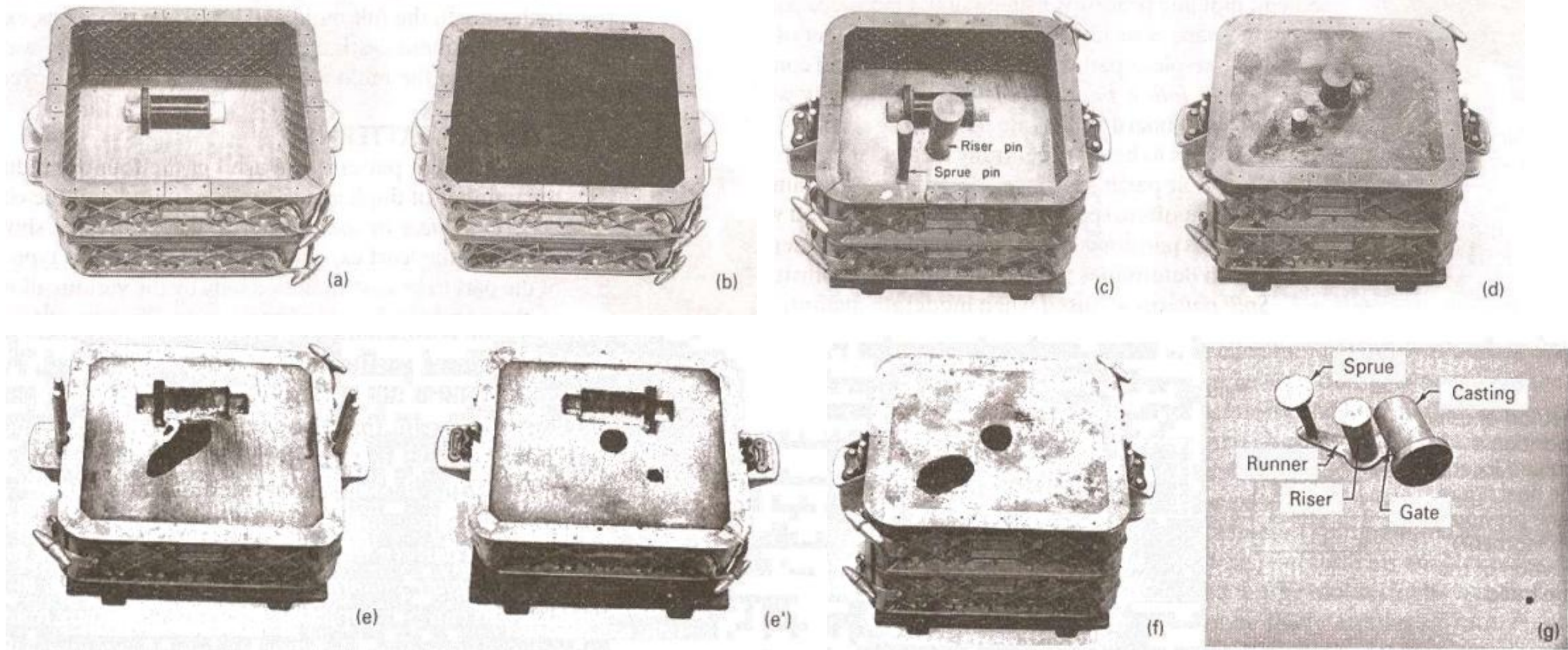
- “Casting is a manufacturing process that involves **pouring of molten metal** into a **mould cavity** where, upon cooling down and **solidification**, it takes the shape of the cavity”



DIFFERENT TYPES OF CASTING PROCESSES

- Expendable mould-type casting processes
 - Sand casting
 - Shell mould casting
 - Plaster mould casting
 - Ceramic mould casting
- Permanent mould-type casting processes
 - Permanent mould casting
 - Vacuum casting
 - Slush casting
 - Pressure casting
 - Die casting
 - Centrifugal casting
 - Squeeze casting

SAND CASTING



FORMING

FORMING

- “Changing the shape of an existing solid body”



Kalpakjian, S., & Schmid, S. R. (2006). Manufacturing Engineering and Technology (6th ed.). Pearson Education South Asia
GIF courtesy: https://commons.wikimedia.org/wiki/File:Blacksmith_Hammer_Strikes_Animation_Loop.gif

DIFFERENT TYPES OF FORMING PROCESSES

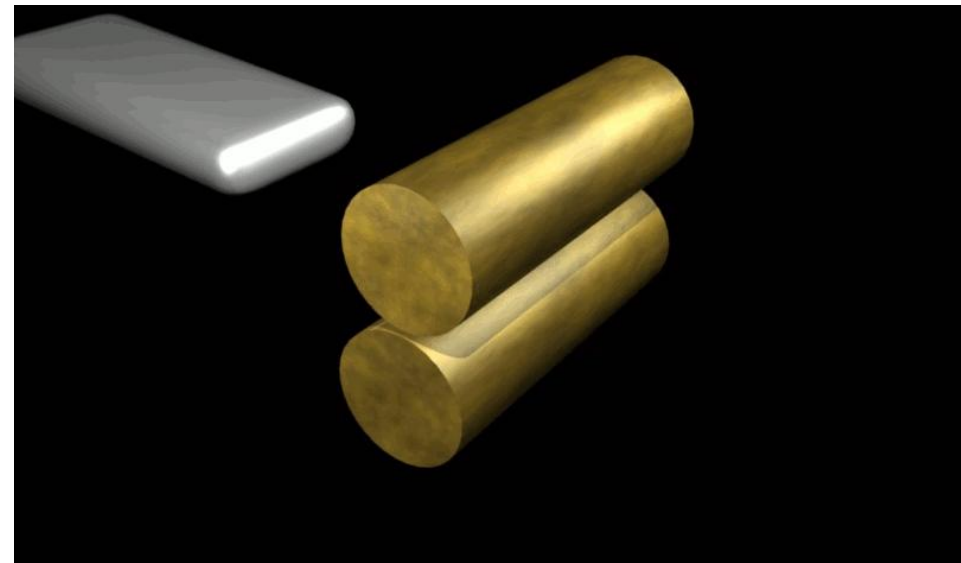
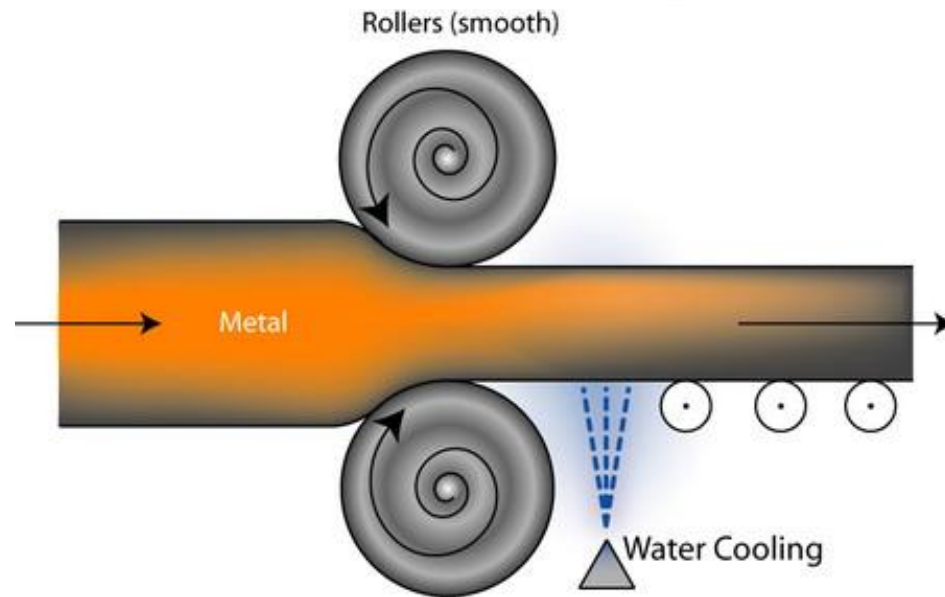
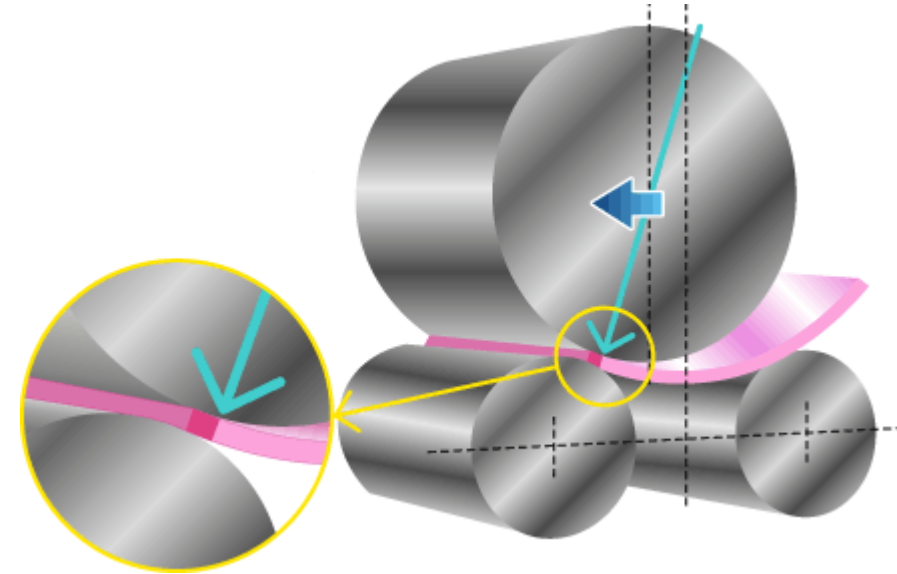
Rolling

Forging

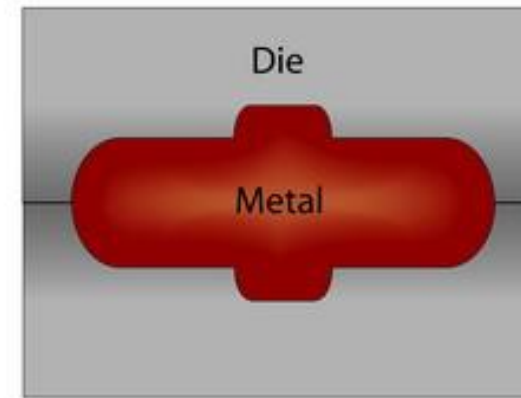
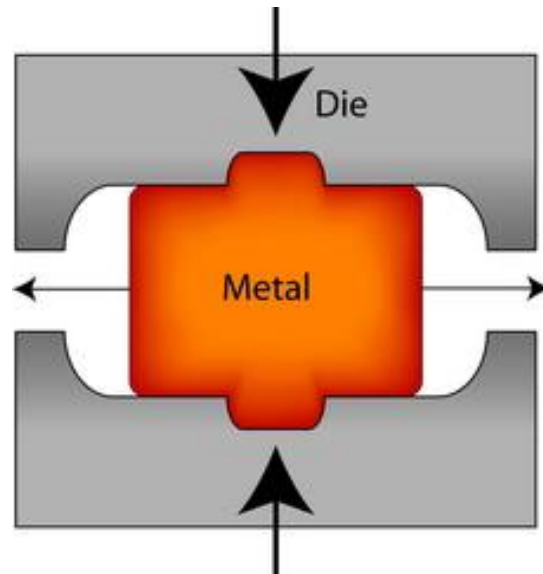
Extrusion

Drawing

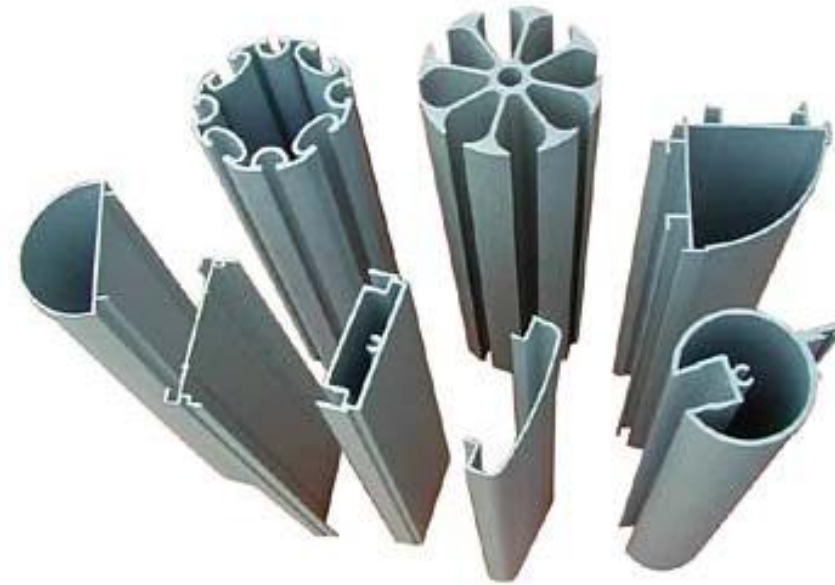
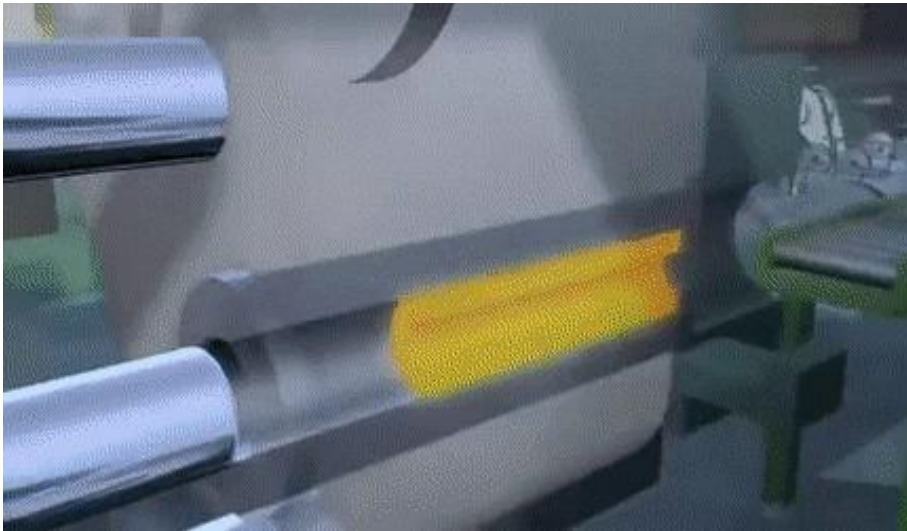
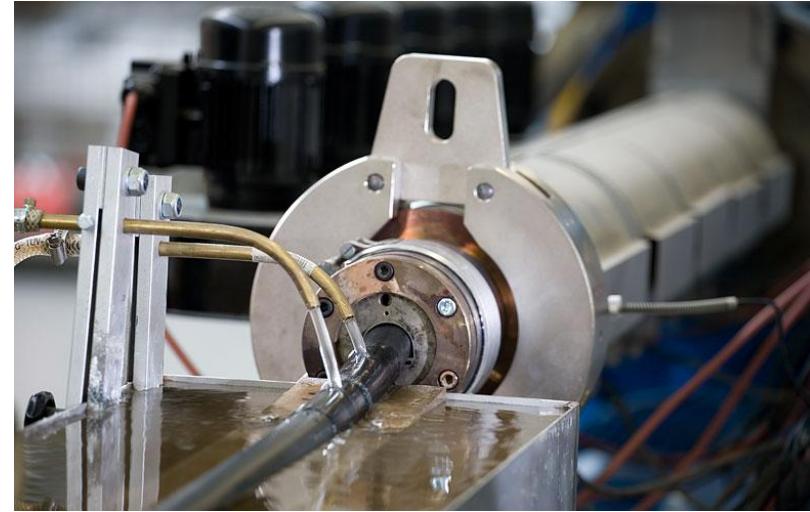
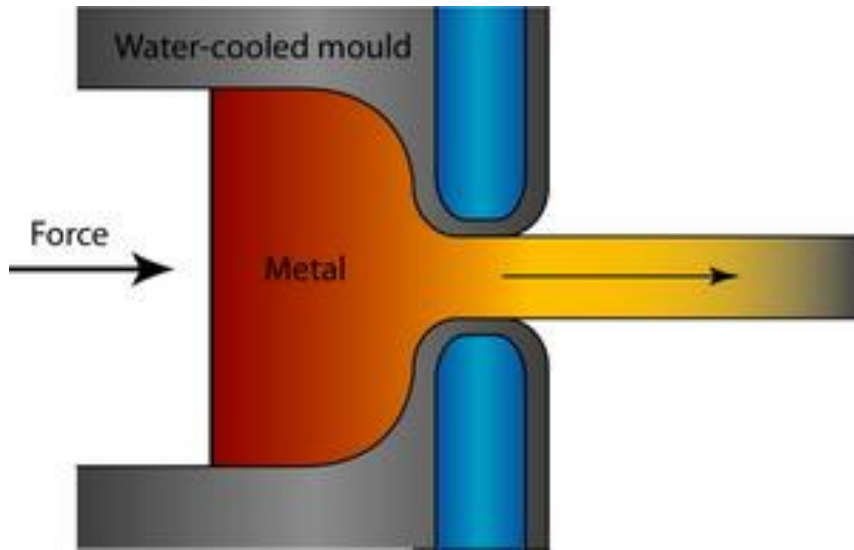
ROLLING



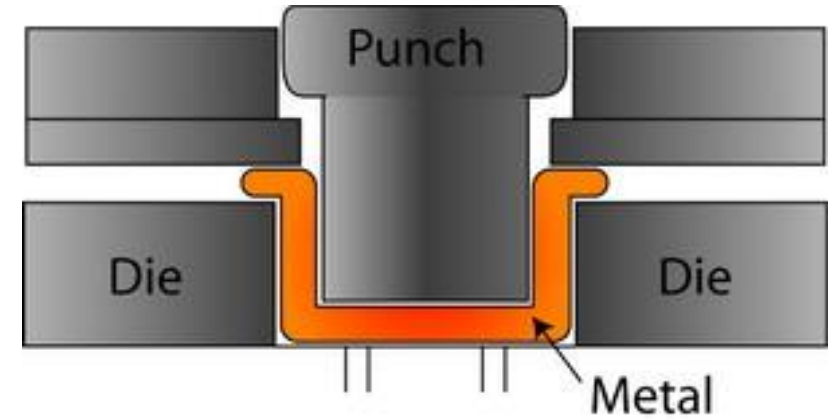
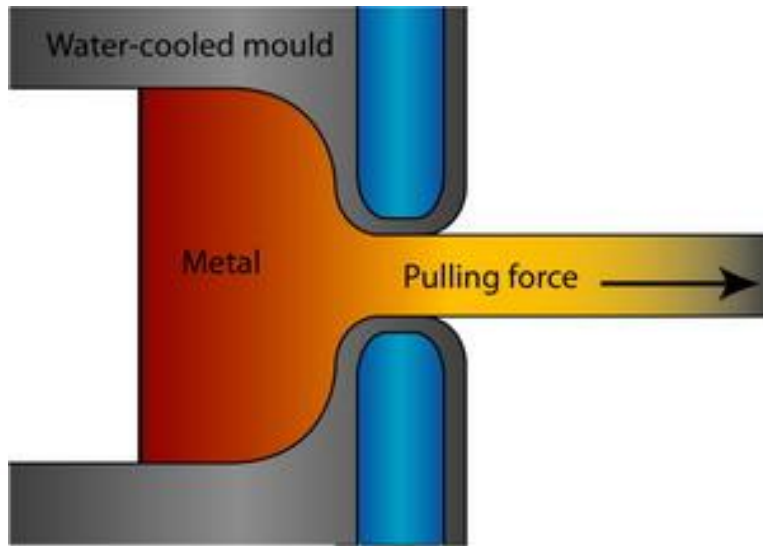
FORGING



EXTRUSION



DRAWING



METAL WIRE



MACHINING

MACHINING

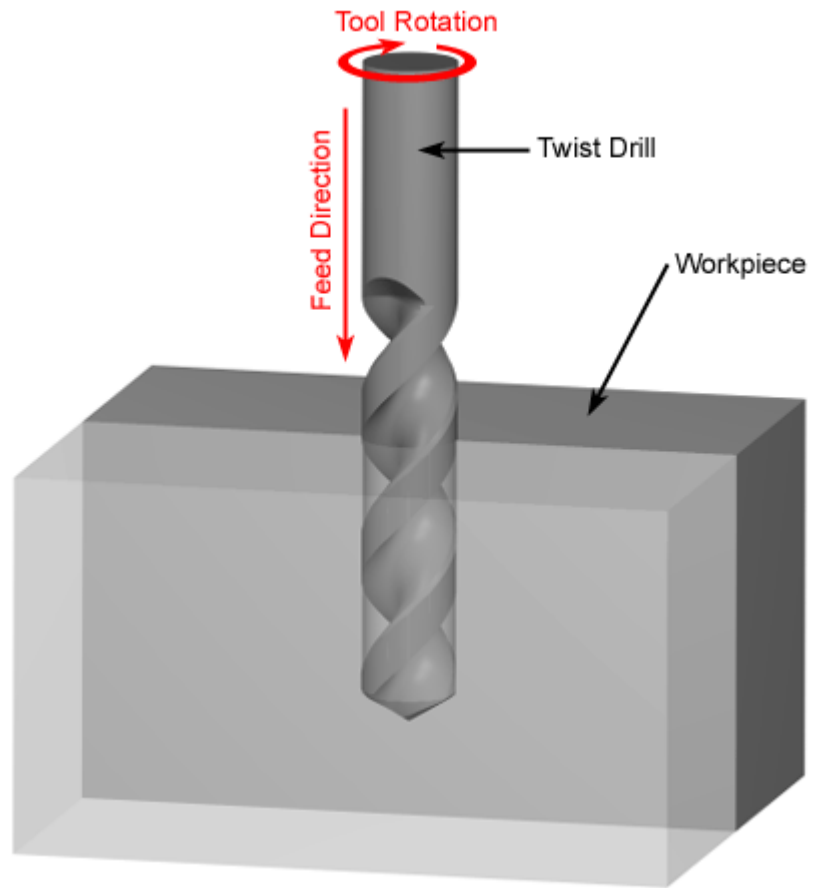
- Machining is a process of **removing material** from a workpiece **to obtain the desired shape** by **producing chips**



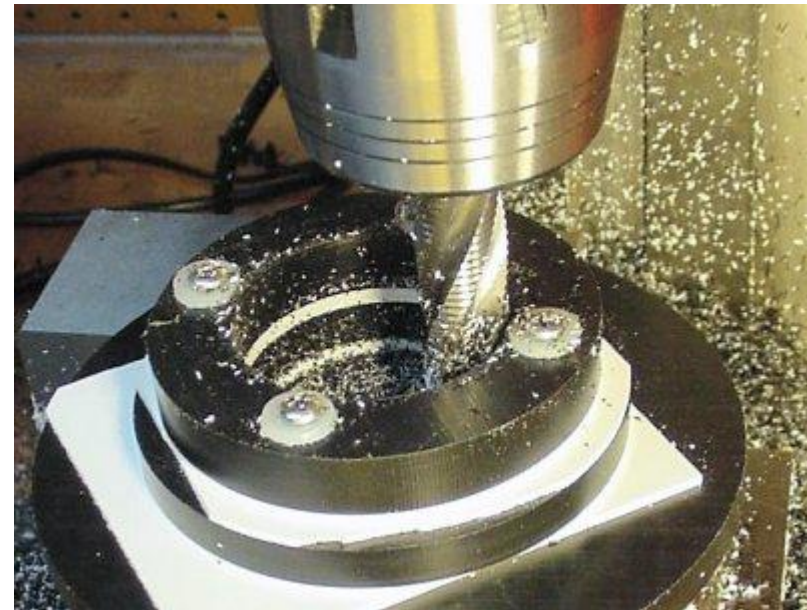
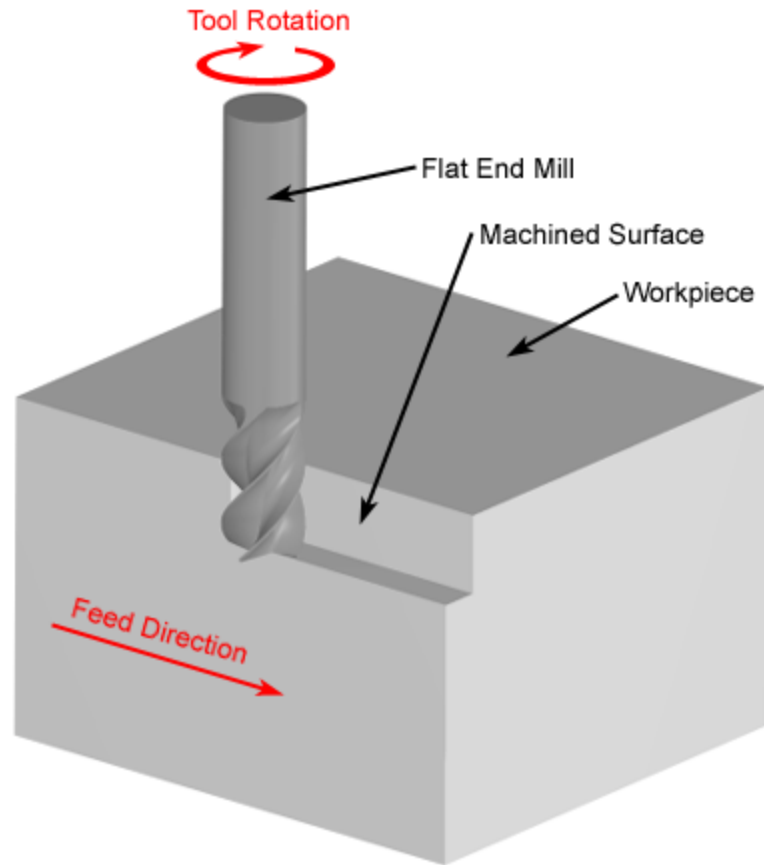
DIFFERENT TYPES OF MACHINING PROCESSES

- Conventional
 - Drilling
 - Milling
 - Turning
 - Tapping
 - Sawing
- Non-conventional
 - Electro Discharge machining (EDM)
 - Water Jet machining (WJM)
 - Laser Beam Machining (LBM)
 - Chemical Machining (CM)
 - Ultrasonic Machining (UM)

DRILLING



MILLING



TURNING

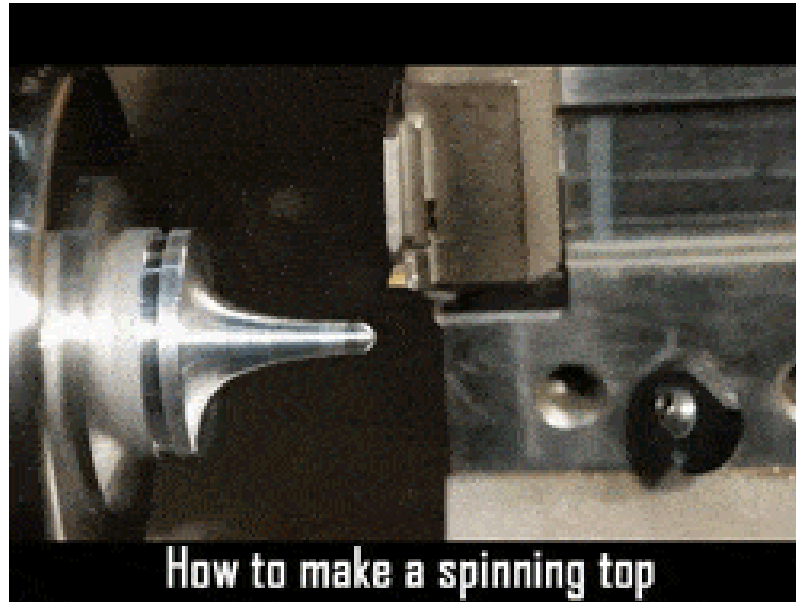
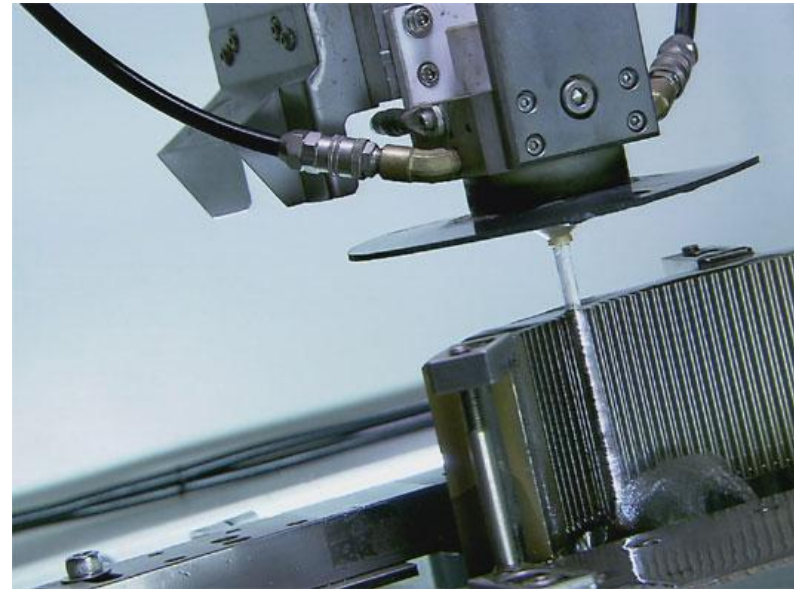
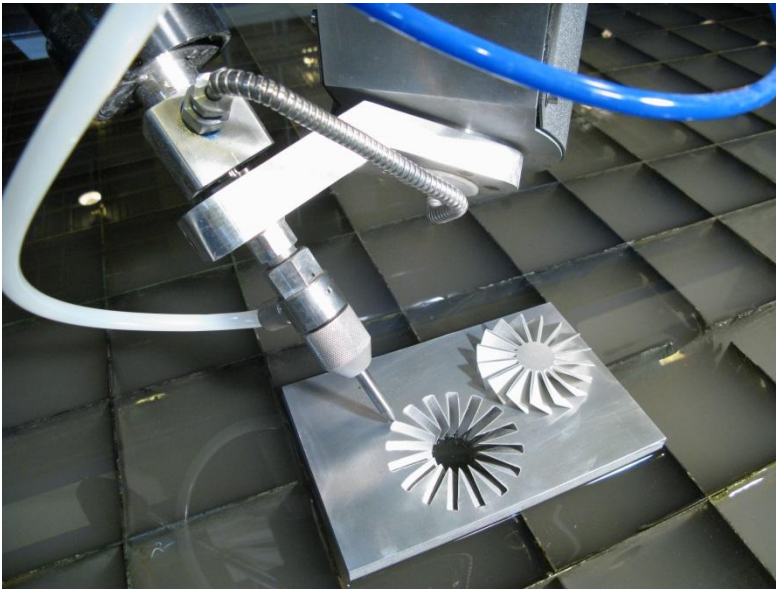


Image courtesy: Internet

05-Mar-26

NON-CONVENTIONAL MACHINING



JOINING

JOINING

- **“Putting parts together to make a single product or putting components together to make an assembly”**



DIFFERENT TYPES OF JOINING PROCESSES

Welding

**Adhesive
bonding**

**Mechanical
fastening**

WELDING

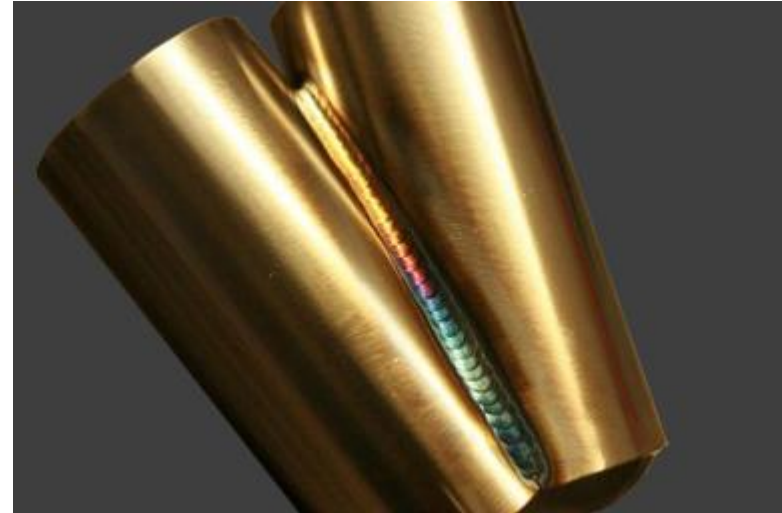


Image courtesy: Internet

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ADHESIVE BONDING

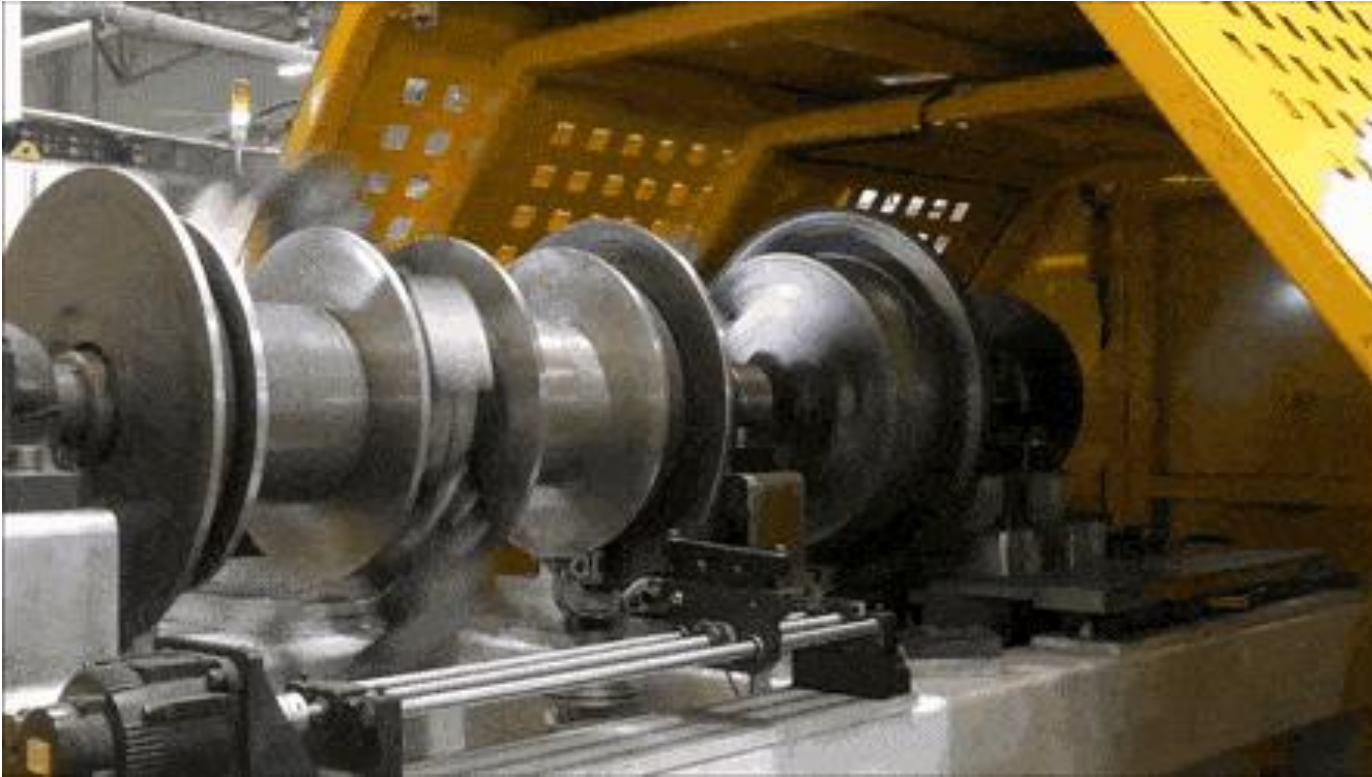


MECHANICAL FASTENING



Image courtesy: Internet

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THE END